Truncated Distribution

1 Definition

Let X be a RV with DF $F(x)$. Let $\mathcal{X}$ be the sample space of $x, 1e.\ p[x \in \mathcal{X}] = 1$. Let, $A = (a, b] \subseteq \mathcal{X}$. Then, the CDF of X over A is,

$$\begin{aligned} G(x) &= P[X \leqslant x \mid x \in A] \\ &= \frac{P[X \leqslant x, a < X \leqslant b]}{P[a < X \leqslant b]} \\ &= \begin{cases} 0, & \text{if } x \leqslant a \\ \frac{P[a < X \leqslant x]}{P[a \leqslant X \leqslant b]}, & a < x < b \\ 1, & x > b \end{cases} = \begin{cases} 0, & x \leqslant a \\ \frac{F(x)-F(a)}{F(b)-F(a)}, & a < x \leqslant b \\ 1, & x > b \end{cases} \end{aligned}$$

Discrete Case

Let $f(x)$ be the PMF of X, the PMF of the truncated distribution of over A is,

$$\begin{aligned} f(x) &= G(x) - G(x-) \\ &= \frac{F(x) - F(a)}{F(b) - F(a)} - \frac{F(x-) - F(a)}{F(b) - F(a)} \\ &= \frac{F(x) - F(x-)}{F(b) - F(a)} = \begin{cases} \frac{f(x)}{P[x \in A]}, & x \in A \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

Continuous case

Let $f(x)$ be the PDF of X, the PDF of X over A is,

$$g(x) = G'(x) = \begin{cases} \frac{f(x)}{P[x \in A)} & x \in A \\ 0 & \text{otherwise} \end{cases}$$

Remark:

$$\int_A f_X(x)dx = P[x \in A] \quad \Rightarrow \int_A \frac{f(x)}{P[x \in A]}dx = 1.$$

So, it is satisfying the condition to be a PDF.

1. In a sequence of independent Bernoulli trail someone recorded the no of success in n trails but forgot to record the outcome with no success. What will be the probability distribution of his counting? Also obtain mean and variance.

Let X be the number of success in n independent Bernoulli trails. $X \sim \text{Bin}(n, p)$.
The PMF of X is,

$$f_X(x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n; x = \{0, 1, 2, \ldots n\}; A = \{1, 2, \ldots n\}$$

$$P[x \in A] = P[X \neq 0] = 1 - P[X = 0] = 1 - q^n$$

The PMF of X over $A = \frac{f(x)}{P[x\in A]} = \frac{\binom{n}{x}p^x q^{n-x}}{(1-q^n)}, \quad x = 1, 2, \ldots n.$

$$\text{Mean: } E(X \mid A) = \sum_{x=1}^{n} x f(x) = \sum_{x=1}^{n} \frac{x\binom{n}{x}p^x q^{n-x}}{(1-q^n)} = \frac{1}{1-q^n}\sum_{x=0}^{n} x\binom{n}{x}p^x q^{n-x} = \frac{np}{(1-q^n)}$$

$$E(X(X-1) \mid A) = \frac{n(n-1)p^2}{(1-q^n)}$$

$$\text{Var}(X) = \frac{n(n-1)p^2}{(1-q^n)} - \left\{\frac{np}{1-q^n}\right\}^2 + \frac{np}{(1-q^n)} = \frac{np}{(1-q^n)^2}\left(1-p+pq^n-npq^n-q^n\right)$$

2. Suppose we are interested to record the dust particle in a certain volume of Air. Data is recorded only when it contains at least one dust particle. What is the distribution of counted no of dust particle in a certain volume. Also derive the mean and variance.

Let X be the number of dust particle in a certain volume. $X \sim$ Poi (λ)
The PMF of X is, $f_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, \ldots; \lambda > 0$

$$\mathcal{X} = \{0, 1, 2, \ldots, \} A = \{1, 2, \ldots\}$$

$P[x \in A] = P[X \neq 0] = 1 - P[X = 0] = 1 - e^{-\lambda}$
The PMF of X over

$$A = \frac{f(x)}{P[x \in A]} = \frac{e^{-\lambda}x^x/x!}{1-e^{-\lambda}} = \frac{e^{-\lambda}\lambda^x}{(1-e^{-\lambda})\,x!}$$

$$E(X \mid A) = \sum_{x=1}^{\infty} \frac{xe^{-\lambda}\lambda^x}{(1-e^{-\lambda})\,x!} = \frac{e^{-\lambda}\lambda}{(1-e^{-\lambda})}\sum_{x} \frac{\lambda^{x-1}}{(x-1)!} = \frac{\lambda}{(1-e^{-\lambda})}e^{-\lambda}e^{\lambda} = \frac{\lambda}{(1-e^{-\lambda})}$$

$$E(X(X-1) \mid A) = \sum_{x} \frac{x(x-1)e^{-\lambda}\lambda^x}{(1-e^{-\lambda})\,x!} = \frac{e^{-\lambda}\lambda^2}{(1-e^{-\lambda})} \quad \sum_{\lambda} \frac{\lambda^x - 2}{(x-2)!} = \frac{\lambda^2}{(1-e^{-\lambda})}$$

$$V(X) = \frac{\lambda^2}{(1-e^{-\lambda})} + \frac{\lambda}{(1-e^{-\lambda})} - \frac{\lambda^2}{(1-e^{-\lambda})} = \frac{\lambda}{(1-e^{-\lambda})}$$

3. Let X be an exponential distribution with mean $\frac{1}{\theta}$. Find the mean and variance of X truncated below μ.

The PDF of $X, f_X(x) = \begin{cases} \theta e^{-\theta x}, & x \geqslant 0 \\ 0, & 0.w \end{cases}$

$$x = (0, \infty)$$
$$A = [\mu, \infty)$$

The PDF of the truncated distribution of X over A is , $g(x) = \begin{cases} \frac{f(x)}{P(x\in A)}, & x \in A \\ 0, & \text{ow} \end{cases}$

$$P(x \in A) = P[X \geqslant \mu] = \int_{\mu}^{\infty} \theta e^{-\theta x}dx = e^{-\theta\mu}g(x) = \begin{cases} \frac{\theta e^{-\theta x}}{e^{-\theta\mu}}, & x \geqslant \mu \\ 0, & \text{otherwise} \end{cases} = \begin{cases} \theta e^{-\theta(x-\mu)}, & x \geqslant \mu \\ 0, & \text{otherwise} \end{cases}$$

It is a shifted Exponential distribution with mean $1/\theta$.

$$E(X \mid A) = \frac{1}{\theta} + \mu, \text{Var}(X \mid A) = \frac{1}{\theta^2}$$

4. Let X follows Cauchy $(0, 1)$. Find the PDF of X over the interval $(-\beta, \beta)$. Find the mean and variance of the truncated distribution if exists. Discuss the use of this truncated distribution.

Yes, it exists. It is a proper integral bounded by $(-\beta, \beta)$.

$$\left[\frac{1}{4}\tan^{-1}x\right]_{-\beta}^{\beta} = \frac{1}{\pi}\left[\tan^{-1}\beta - \tan^{-1}(-\beta)\right]$$
$$= \frac{1}{\pi}\tan^{-1}\frac{\beta+\beta}{1+\beta\cdot(-\beta)} = \frac{1}{\pi}\tan^{-1}\frac{2\beta}{1-\beta^2}$$

$\therefore P(x \in A) = \frac{1}{\pi}\tan^{-1}\frac{2\beta}{1-\beta^2}$.

The PDF of truncated distribution over $(-\beta,\beta) = \frac{\frac{1}{\pi}\tan^{-1}x}{\frac{1}{\pi}\tan^{-1}\frac{2\beta}{1-\beta^2}} = \frac{1}{2\tan^{-1}\beta(1+x^2)}$ if $|x| < \beta$.

$(-\beta, \beta)$

5. $X \sim N\left(\mu, \sigma^2\right)$. Let $A = (a, b)$. Find the mean and variance of the truncated distribution of X over A. In particular, $A = (\mu - k\sigma, \mu + k\sigma)$, show that the truncated distribution is symmetric and hence show that, $E(X \mid A) = \mu_1$, variance γ^2 satisfies,

$$\left(1 - \frac{\gamma^2}{\sigma^2}\right) I(k) = ke^{-k^2/2}$$

$$X \sim N\left(\mu, \sigma^2\right), \text{PDF of } X, f(x) \quad = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right).$$

$$P[x \in A] = P[a < X < b] = P\left[\frac{a-\mu}{\sigma} \leqslant \frac{x-\mu}{\sigma} \leqslant \frac{b-\mu}{\sigma}\right]$$
$$= \Phi(b) - \Phi(a)$$

The PDF of X over A is, $g(x) = \frac{f(x)}{P[x\in A]} = \begin{cases} \frac{\frac{1}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right)}{\Phi(b') - \Phi(a')} & , a < X < b \\ 0, & \text{otherwise} \end{cases}$

$$E\left(\frac{X-\mu}{\sigma}\middle|\, A\right) = \int_A \left(\frac{x-\mu}{\sigma}\right) g(x)dx = \int_a^b \frac{\left(\frac{x-\mu}{\sigma}\right)}{\sigma} \frac{\phi\left(\frac{x-\mu}{\sigma}\right)}{\Phi(b') - \Phi(a')} dx, \quad \frac{x-\mu}{\sigma} = z$$
$$= \int_{a'}^{b'} z \frac{\phi(z)dz}{P(x \in A)}$$
$$= \frac{1}{P(x \in A)}[-\phi(z)]_{a'}^{b'} = \frac{\phi(a') - \phi(b')}{\Phi(b') - \Phi(a')} \to \text{It is the PDF.)}$$
$$E(X) = \mu + \sigma \quad \frac{\phi(a') - \phi(b')}{\Phi(b') - \Phi(a')}$$
$$E\left\{\left(\frac{X-\mu}{\sigma}\right)^2 \middle|\, A\right\} = \frac{1}{P[x \in A]} \int_a^b \left(\frac{x-u}{\sigma}\right)^2 g(x)dx$$
$$= \frac{1}{P[x \in A]} \int_{a'}^{b'} z^2 \phi(z)dz$$
$$= \frac{1}{P[x \in A]} \left[-z\phi(z)|_{a'}^{b'} + \int_{a'}^{b'} \phi(z)dz\right]$$
$$= \frac{a'\phi(a') - b'\phi(b')}{\Phi(b') - \Phi(a')} + 1$$
$$V(X \mid A) = V(X - \mu \mid A) = \sigma^2 \operatorname{Var}\left(\frac{X-\mu}{\sigma}|A\right) = \sigma^2\left[\left(E\left(\frac{X-\mu}{\sigma}\right)^2 | A - E^2\left(\frac{X-\mu}{\sigma}|A\right)\right)\right]$$
$$= \sigma^2\left[\frac{\left(\frac{a-\mu}{\sigma}\right)\phi\left(\frac{a-\mu}{\sigma}\right) - \left(\frac{b-\mu}{\sigma}\right)\phi\left(\frac{b-\mu}{\sigma}\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)} - \left\{\frac{\phi\left(\frac{a-\mu}{\sigma}\right) - \phi\left(\frac{b-\mu}{\sigma}\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)}\right\}^2\right]$$

i) Particular case: $A = (a, \infty)$ (left tail truncation)

$$g(x) = \frac{\frac{1}{\sigma}\phi\left(\frac{x-\mu}{\sigma}\right)}{1 - \Phi\left(\frac{a-\mu}{\sigma}\right)}$$

ii) Particular case: $A = (\mu - k\sigma, \mu + k\sigma), \quad g(x) = \frac{f(x)}{P[x \in A]}$

$$\begin{aligned}
\text{Here, } P[x \in A] &= P[\mu - k\sigma < X < \mu + k\sigma] \\
&= P\left[-k < \frac{X-\mu}{\sigma} < k\right] \\
&= \int_{-k}^{k} \frac{1}{\sqrt{2\pi}} e^{-x/2} dx = \frac{2}{\sqrt{2\pi}} \int_0^k e^{-x^2/2} dx = \sqrt{\frac{2}{\pi}} I_k
\end{aligned}$$

$$g(x) = \frac{\frac{1}{\sigma}\phi\left(\frac{x-u}{\sigma}\right)}{\sqrt{2/\pi} I_k}$$

Note that, $g(\mu - x) = g(\mu + x)\ \forall x$
$\Rightarrow$ The truncated distribution is symmetric about μ.

$$\begin{aligned}
&E(X \mid A) = \mu \\
&\text{Var}(X \mid A) = E\left[(X-\mu)^2 \mid A\right] \\
&= \sigma^2 \int_{\mu-k\sigma}^{\mu+k\sigma} \left(\frac{x-\mu}{\sigma}\right)^2 g(x) dx \\
&= \sigma^2 \int_{-k}^{k} \frac{z^2 \phi(B)}{P[x \in A]} dz \\
&= \frac{2\sigma^2}{\sqrt{\frac{2}{\pi}} I_k} \left[-z\phi(z)|_0^k + \int_0^k \phi(z) dz\right] \\
&= \frac{2\sigma^2}{\sqrt{\frac{2}{\pi}} I_k} \left[-k\phi(k) + \frac{1}{\sqrt{2\pi}} I(k)\right] \\
&= \frac{\sigma^2}{\frac{1}{\sqrt{2\pi}} I(k)} \left\{\frac{-ke^{-k^2/2}}{\sqrt{2\pi}} + \frac{1}{\sqrt{2\pi}} I(k)\right\} \\
&= \sigma^2 \left\{-\frac{ke^{-k^2/2}}{I(k)} + 1\right\} \\
&\left(1 - \frac{v^2}{\sigma^2}\right) I(K) = ke^{-K^2/2} \quad \text{(Proved).}
\end{aligned}$$

As the truncated distribution is symmetric $\delta_3(A) = 0$.
Note. $\mu_4(A) = \frac{\sigma^4}{I(k)} \left\{3I(k) - k^3 e^{-k^2/2} - 3ke^{-k^2/2}\right\}$ (Integrating by parts for 2 times)

$$\begin{aligned}
\mu_4(X \mid A) &= \sigma^4 E\left\{\left(\frac{X-\mu}{\sigma}\right)^4 \middle| A\right\} = \frac{2\sigma^4}{p(X \in A)} \int_0^k z^3 z\Phi(z) dz \\
&= \frac{\sigma^2}{I(k)} \left\{3I(k) - k^3 e^{-k^2/2} - 3ke^{-k^2/2}\right\} \\
\gamma_2(x \mid A) &= \left\{\frac{\mu_4(A)}{\mu_2^2(A)} - 3\right\} = \frac{-ke^{-k^2/2} \left\{\left(k^2 - 3\right) I(k) + 3ke^{-k^2/2}\right\}}{\left\{I(k) - ke^{-k^2/2}\right\} 2} \\
&< 0 \quad \text{if } k > \sqrt{3}
\end{aligned}$$

If $k > \sqrt{3}$, the truncated distribution is pletykurtic.